Allen Zheng

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Education

The University of Texas at Dallas GPA: 3.777 / 4.000

Bachelor of Science in Computer Science *Expected Graduation May 2028*

Honors: National Merit Scholars Program, Computing Scholars Honors Program, Hobson Wildenthal Honors College

Courses: Data Structures & Algorithms, Software Engineering, Computer Architecture, Databases, Machine Learning

Experience

Project Manager | Association for Computing Machinery at UTD May 2026 – Present

Leadership, Agile, System Architecture, Technical Mentorship *Richardson, TX*

- Lead **4** new developers through **5** two-week **Agile** sprints building **Echo**, an AI-narrated audiobook mobile app

Software Engineering Intern | Wink Jan 2026 – Aug 2026

Kotlin, Android, Biometrics, Security, Networking Protocols, Performance Optimization, Azure *Plano, TX*

- Built a **Kotlin**/Android app wirelessly linking POS registers to the biometric SDK, authenticating under **800ms**
- Optimized performance by migrating the codebase from **Flutter** to native **Kotlin**, cutting **RAM usage by 50%**
- Unified **Wi-Fi**, **Bluetooth RFCOMM**, and **USB** register-terminal links behind one transport-agnostic interface
- Secured pairing with **mutual TLS** and hardware-backed keys, replacing 64-char fingerprints with an 8-digit code

Development Officer | Association for Computing Machinery at UTD Dec 2025 – Present

Golang, AWS EC2, Cloudflare, GCP, Firebase, Google OAuth, React, TypeScript, SQLite *Richardson, TX*

- Built **REST** endpoints in **Golang** serving UTD coursebook, grade, and professor data from **Firebase Firestore**
- Deployed the API on **AWS EC2** and **Cloudflare**, architecting **Google OAuth** login and API key management
- Fixed a **4-year-old** bug in ACM's **utdgrades.com** that dropped all F grades, skewing class sizes and averages
- Built section sorting/filtering by mean GPA and median grade in **React**, queried in-browser over **SQLite** via **sql.js**

Projects and Awards

Fitts (Meta Global AI Hackathon) | *TypeScript, Electron, React, Playwright, Android, Muse* Oct 2026 – Nov 2026

- Solo-built **fitts.allenzheng.com**, a desktop **AI agent** that audits web and Android apps and fixes their bugs
- Found **7x** as many planted issues as **axe** (**28/33**) at $0.15 a run, and **3x** as many on a blind held-out app
- Built an autonomous fix loop that verified **13** of **16** queued fixes in a **git worktree**, rolling back any failures

Reagent (HackTX NMC2 Track 3rd Place) | *Python, FastAPI, PostgreSQL, LangGraph, MCP, Next.js* Oct 2026

- Architected an enterprise **AI-agent platform** where every agent inherits org-wide models, tools, and guardrails
- Designed a **7**-stage **CI/CD** pipeline gating every agent release on security review, human-approved tests, and evals
- Built a statistically rigorous **eval engine** scoring repeated sandboxed runs with **95%** Wilson confidence intervals

Open Team Sheets | *TypeScript, Next.js, React, PostgreSQL, Supabase, GitHub Actions* Sept 2026 – Present

- Solo-built **openteamsheets.com**, an open-source **Next.js**/**PostgreSQL** site searching **38K+** Pokémon teams
- Cut box-search queries from **73–88ms** to **24–29ms** with **1024-bit bitset** masks, ending 3s+ production timeouts
- Set up **CI** gating every PR on format, lint, typecheck, build, **186 Vitest** unit tests, and **pgTAP** database tests

MindCraft (HackAI Track Winner) | *React, TypeScript, TensorFlow, Random Forest, Gemini API* Mar 2026

- Built a full-stack **ML** platform in **React**/**TypeScript** taking users from raw data to a live model in **5 minutes**
- Designed ML pipeline supporting tabular/image data using **Random Forest** + **MobileNetV2** transfer learning

Skills

Programming Languages: Python, Java, JavaScript/TypeScript, C, C#, C++, Kotlin, Swift, Golang, SQL
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Express, Flask, Electron, Flutter, .NET, TensorFlow
AI & Agents: LangGraph, MCP, LLM Evaluation, LLM-as-Judge, RAG, AWS Bedrock, OpenRouter, Claude Code
Databases: PostgreSQL, Supabase, MySQL, MongoDB, SQLite, Firebase Firestore, DynamoDB, AWS S3
Developer Tools: AWS, Azure, Google Cloud (GCP), Docker, Git, GitHub Actions, Vercel, Unix/Linux
General: System Design/Architecture, Object-Oriented Design, REST APIs, CI/CD, Unit Testing, Code Review, Agile